

NEUROLOGICALLY MOTIVATED SELF-SUPERVISED LEARNING FOR BILATERAL GAIT GEOMETRY

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INTRODUCTION

Human-aware robots often begin with a perception module that summarizes a person’s pose and motion. Before such a summary is used for prediction or interaction, it should be tested for geometric information relevant to the task. We study a small but concrete case: a reflected skeleton swaps anatomical sides and reverses a left–right speed contrast. Stroke and Parkinson’s disease can alter gait asymmetry [1], which motivates the geometry; our target remains a pose-derived movement score rather than a clinical label. We test whether masked joint-embedding predictive architecture (JEPA) training makes this score easier to recover.

MATERIALS AND METHODS

GAVD contributes 625 quality-checked clips from 93 source videos [2]. We process each clip independently, retaining its 33 landmarks in a 64-time-step, three-coordinate array. Four consecutive steps form one patch, preserving a 16-by-33 time-by-joint layout. The online encoder sees visible patches, a predictor estimates hidden-patch features, and an exponential-moving-average teacher provides targets [3]. No movement score or affected-side label enters encoder training.

The target is the mean normalized difference of left and right median speeds across five paired joints. Five outer folds hold out complete source videos; every clip from a held-out source remains outside encoder training, readout fitting, and readout tuning. Five seeds repeat the matched comparison of each trained encoder with its own starting weights. We resample whole sources for paired intervals while holding fitted models fixed.

RESULTS AND DISCUSSION

The initial encoder reached $R^2 = 0.223$ with a motion-sensitive frozen readout. The five trained teacher encoders reached 0.101–0.114; their paired differences from initialization ranged from -0.109 to -0.122 . Yet trained predictors preferred correct-clip teacher features to features from another source in 375 of 375 diagnostic checks, compared with 33 of 75 checks before training. Feature correspondence therefore improved without improving the tested bilateral-motion estimate.

This is a useful failure mode for human-aware perception, but its reach is limited. The expanded readout also adds temporal and missingness statistics, and preprocessing changes the target calculation. The evidence identifies a shortfall in this training-and-readout pipeline rather than proving that trained features lack all geometric motion information.

CONCLUSIONS

Masked feature prediction can capture clip-related structure while making a relevant bilateral motion quantity less accessible to a frozen linear readout. The result motivates explicit task-level checks before using self-supervised articulated-motion features downstream. It does not claim robotic performance or control benefit.

REFERENCES

- [1] Lewek MD et al. Gait Posture 31:256–260, 2010.
- [2] Ranjan R et al. IEEE Access 13:45321–45339, 2025.
- [3] Assran M et al. arXiv:2301.08243, 2023.